AIM:

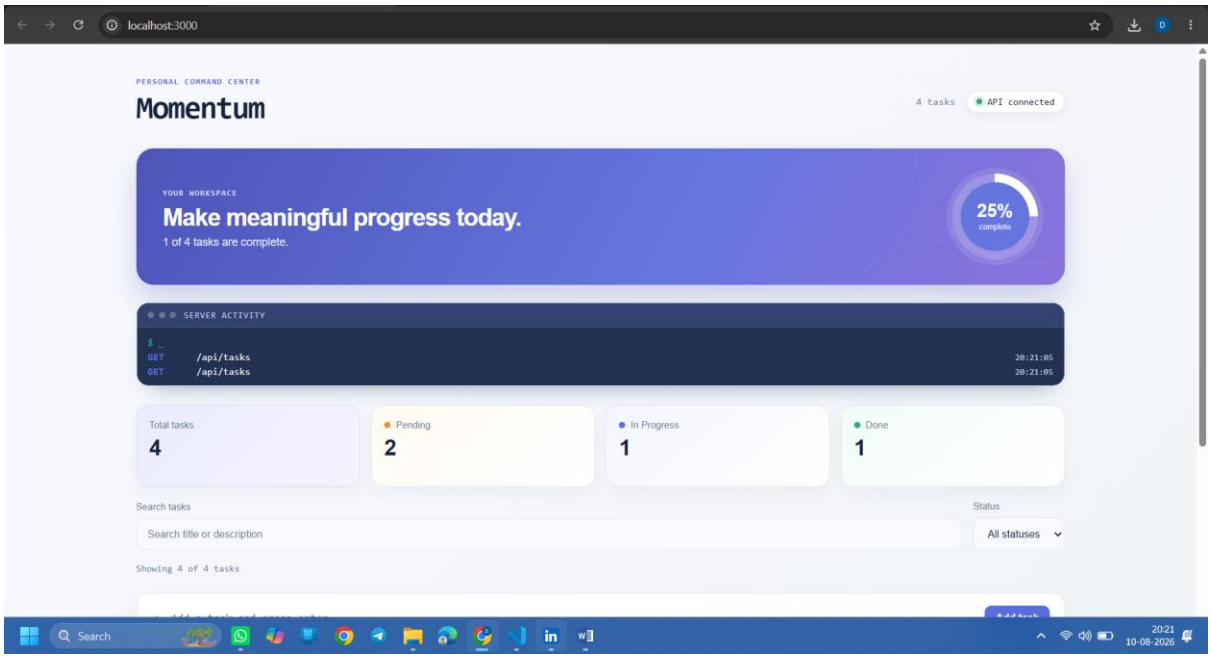
To design and implement a RESTful backend API using Node.js and Express.js with complete CRUD operations and a middleware pipeline for request logging and centralized error handling.

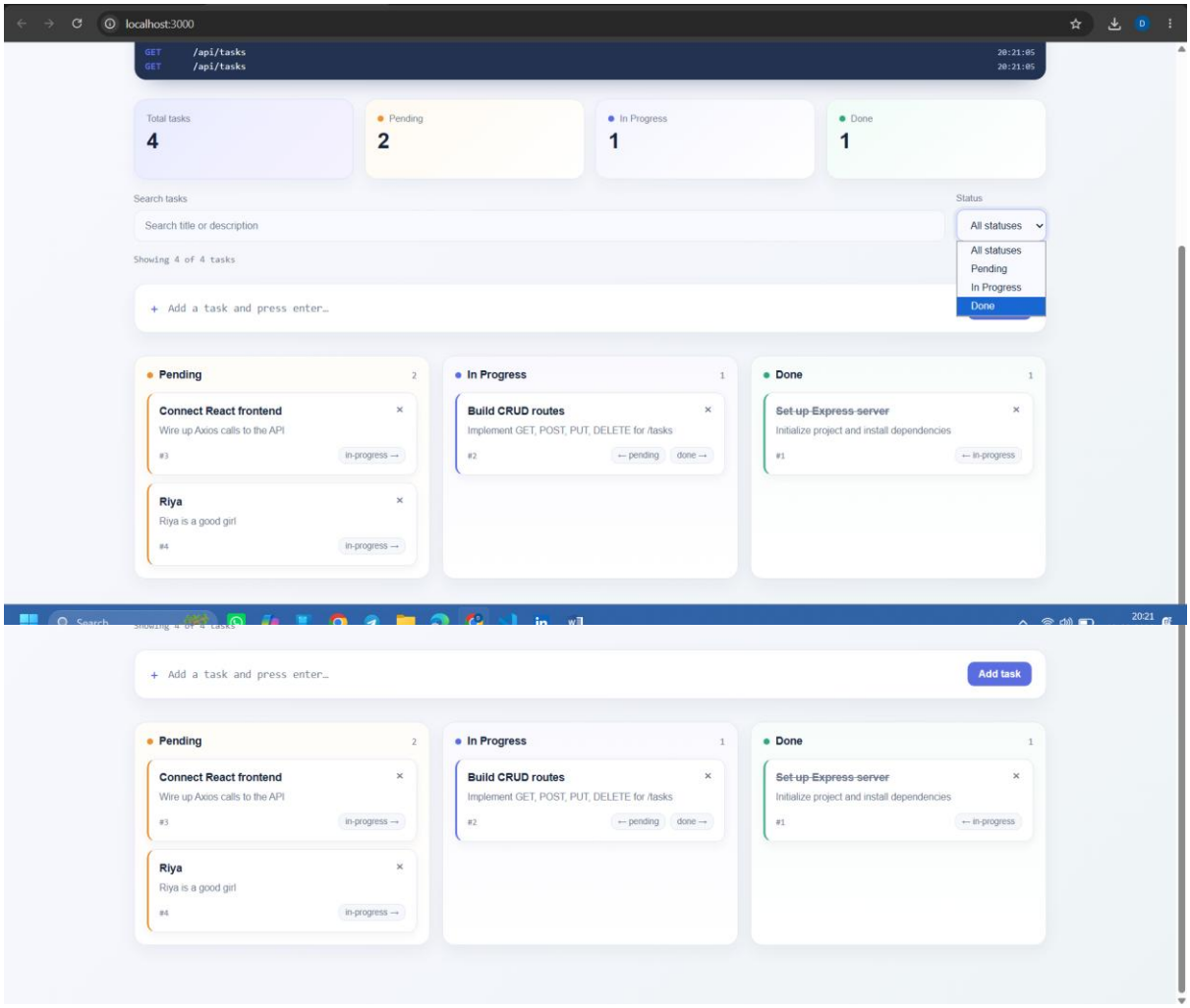
CONTENT:

After completing this practical, I will be able to design and implement a RESTful backend server using Node.js and Express.js. I will understand how to create REST API endpoints for performing CRUD operations such as creating, reading, updating, and deleting tasks using HTTP methods including GET, POST, PUT, and DELETE. I will also learn how Express middleware works within the request-response lifecycle and how to implement custom middleware for request logging and centralized error handling. Additionally, I will understand how to use appropriate HTTP status codes such as 200, 201, 404, and 500 to provide meaningful responses to clients. I will be able to handle invalid requests and server-side errors gracefully and test the implemented API endpoints using tools such as Postman or Thunder Client. This practical will help me develop structured, reliable, and maintainable backend applications using Node.js and Express.js.

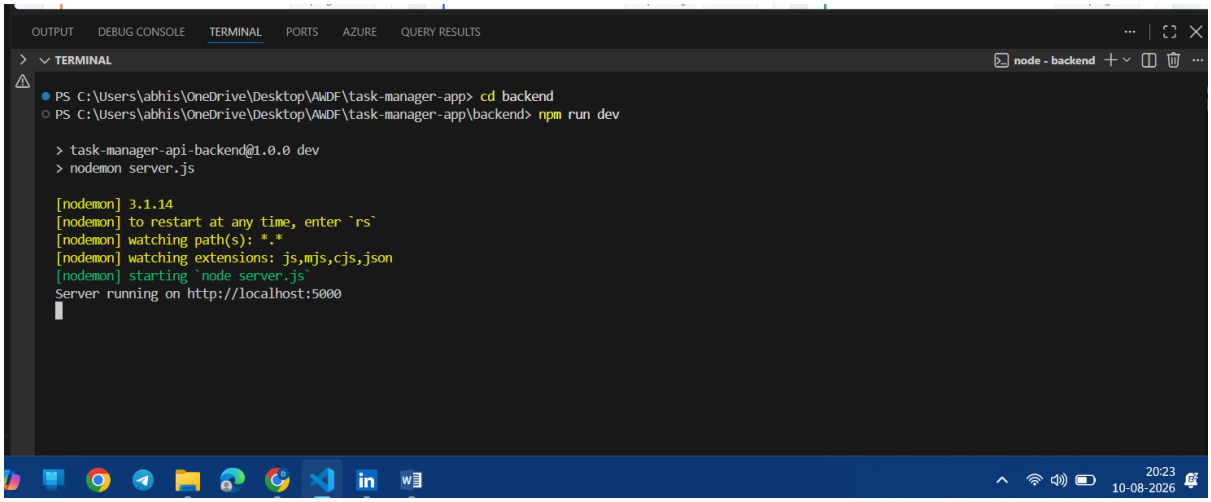
Screen Shot:

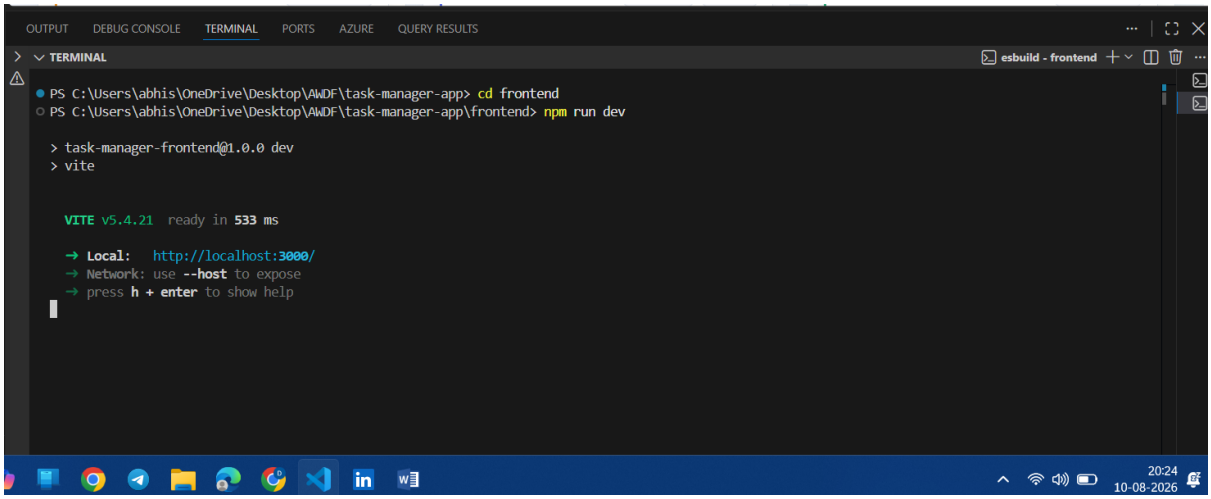
Website related:





Terminal related:





Conclusion

This practical demonstrates the development of a RESTful backend server using Node.js and Express.js for a Task Management system. It implements complete CRUD operations that allow users to create, retrieve, update, and delete tasks through REST API endpoints. The project also demonstrates the use of Express middleware for request logging, JSON data handling, and centralized error management. Appropriate HTTP methods and status codes are used to ensure consistent and meaningful communication between the client and server. The API is tested using tools such as Postman or Thunder Client to verify the functionality and correctness of each endpoint. Overall, this practical provides hands-on experience in building structured, reliable, and maintainable backend services using Node.js and Express.js.